

$$\text{At } Y, v_Y^2 = u^2 + 2a(s/2) = u^2 + as = u^2 + \frac{1}{2}(v^2 - u^2) = \frac{1}{2}(u^2 + v^2)$$

- 4 **C** During fall in air, weight of ball - viscous force = ma

Initially the ball is accelerating at a decreasing rate since viscous force increases with speed.

At terminal speed, weight of ball - viscous force, i.e. $mg = kv$

If mass m is less, terminal speed is less.

The gradient of $s - t$ graph gives the velocity v . answer is C.

- 5 **D** Let F be the upward force by $3m$ on the system of $(m + 5m)$.